

Panel	Model	Climate Model	AIC	Akaike Weight	Deviance Explained	R ² Adj	Significant terms (term-level p) ¹	Significant contrasts vs. reference	EDF Terms
Punta Maroma	M4_av	Thermal-disturbance model (intensity)	-55.45	0.81	0.825	0.715	s(Hur100_av) (p=<1e-04)	n/a (no categorical terms)	s(age) (EDF=0.83); s(GMHT_CPS0_30N) (EDF=1.12); s($\delta T, ^\circ C$) (EDF=0.52); s(Hur100_av) (EDF=4.07)
Florida	M4_av	Thermal-disturbance model (intensity)	-14.00	0.32	0.287	0.177	s(Hur100_av) (p=0.0092)	n/a (no categorical terms)	s(age) (EDF=0); s(GMHT_CPS0_30N) (EDF=1.49); s($\delta T, ^\circ C$) (EDF=0); s(Hur100_av) (EDF=1.59)
Belize	M6_fqcy	Integrated climate model (frequency)	-21.03	0.64	0.790	0.600	Thermal (p=0.0017); s(GMHT_CPS0_30N) (p=0.033); s(Hur100_fqcy) (p=0.032)	Hydrovariable (1.16, p=0.039); Thermal CWP & MWA (-1.77, p=0.024); ThermalLIA (-2.39, p=0.0033)	s(age) (EDF=0.54); s(GMHT_CPS0_30N) (EDF=1.71); s($\delta T, ^\circ C$) (EDF=0); s(Hur100_fqcy) (EDF=1.55)
Saint Croix	M4_fqcy	Thermal-disturbance model (frequency)	-10.51	0.31	0.583	0.429	s($\delta T, ^\circ C$) (p=0.0092); s(Hur100_fqcy) (p=0.032)	n/a (no categorical terms)	s(age) (EDF=0.16); s(GMHT_CPS0_30N) (EDF=0); s($\delta T, ^\circ C$) (EDF=1.55); s(Hur100_fqcy) (EDF=0.78)
Regional	M6_av	Integrated climate model (intensity)	-25.86	0.30	0.439	0.269	Thermal (p=0.025)	Thermal LIA (-1.96, p=0.0032)	s(age) (EDF=2.75); s(GMHT_CPS0_30N) (EDF=0.6); s($\delta T, ^\circ C$) (EDF=0); s(Hur100_av) (EDF=0)

¹Term-level p: joint Wald test of the whole term (all factor levels together; the spline for continuous predictors); s(age) excluded. Contrasts are relative to the reference levels warmer / wet / low. Rank-deficient candidates were excluded before model selection.